

Practical 6

* Aim:- For a given set of relation schemes, create tables and perform the following Simple Queries, Simple Queries with Aggregate functions, Queries with Aggregate functions (group by & having clause).

* Theory:- A table Student2 is created to store student details such as Student ID, Name, Department, and marks.

The following SQL operations are performed:-

(i) Table creation:- The create table command is used to create the structure of the table with appropriate data types.

Ex:- create table Student2 (StudID INT, Name varchar(20), Dept Varchar(20), Marks INT);

(ii) Multiple records were inserted to add student records in the table using the Insert into command.

Ex:- Insert into student2 values (1, "Rishabh Dwivedi", "MCA", 75);

(iii) Aggregate Functions:- Aggregate functions perform calculations on multiple rows and return a single result.

(a) Count :- Counts the total numbers of

records

Ex:- Select count(*) as total-students from Student2;

⑥ AVG function:- calculates the average of the marks column.

Ex:- Select AVG(Marks) As average-marks from Student2;

⑦ max function:- Returns the highest mark obtained by students

Ex:- select max(Marks) as highest-marks from Student2;

⑧ min function:- Returns the minimum marks from the table.

Ex:- select min(Marks) as lowest marks from Student2;

⑨ Sum function:- Calculates the total of all marks

Ex:- select sum(Marks) as total-marks from Student2;

* group By Clause:- This clause is used to arrange rows into groups based on similar values in a column it is mostly used with aggregate functions.

Ex:- select Dept, avg(Marks) as Avg-marks from Student2 group by Dept;

This query groups students according to their department and calculates the average marks of each department.

* Having Clause :- The having clause is used to apply conditions on grouped data. It works similarly to the where clause, but having is specifically used with aggregate functions.

Ex:- select Dept, Avg(Marks) as Avg-Marks from Student2 group by dept having Avg(Marks) > 80;
This query displays only those departments whose average marks are greater than 80.

* Conclusion :- In this practical, SQL queries were performed on the Student2 table using simple queries, aggregate functions, group by, and having clauses. These operations helped in retrieving and analyzing student data efficiently and improved understanding of database query operation in SQL.